AIDEN ZHANG

COMPUTER ENGINEERING AT THE UNIVERSITY OF WATERLOO

VIEW INTERACTIVE PORTFOLIO

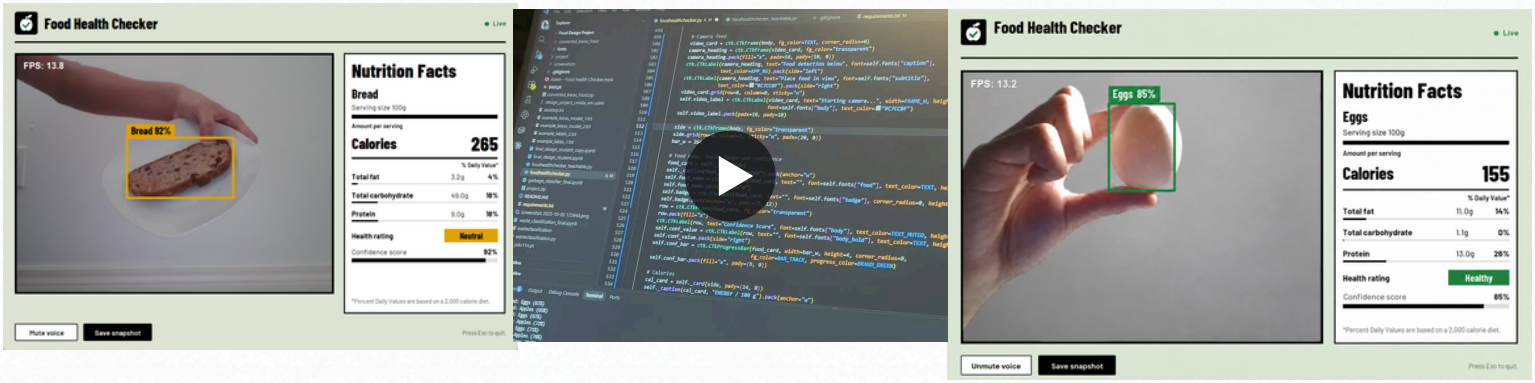
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COMPUTER VISION FOOD RECOGNITION AND NUTRITIONAL ANALYSIS SYSTEM
PYTHON, PYTORCH, YOLO



What:

- A live webcam app that recognizes five foods, draws a box around each one, and shows its nutrition facts beside the camera feed
- Reads the detected food and its nutrition aloud for hands-free use

How:

- Fine-tuned a YOLO11 detector on 245 hand-labelled images
- Built a three-thread pipeline: camera and detection, GUI, and speech, so slow frames never freeze the window
- Filtered detections at 50% confidence and held boxes for 8 missed frames to stop flicker

Result:

- 98% mAP50 on a held-out validation set
- Averaged 13.5 FPS across 300 frames, including capture, inference and drawing
- Used in a five-month pilot with six participants

DRONE BEHAVIOUR TREE TELEMETRY SYSTEM - PYTHON, ROS 2, DOCKER



What:

- A debugging link that shows which behaviour tree step the drone's autonomy is running, even when there is no LTE connection to the drone
- Sends that step over MAVLink STATUSTEXT, the same link the pilots already use, by compressing long behaviour tree names into short IDs that fit its low-bandwidth, 50-character messages

How:

- Read behaviour tree snapshots every 0.5 s and selected the deepest RUNNING node
- Numbered every node and sent only a short ID (S,8 instead of FlyToWaypoint) over MAVLink STATUSTEXT through MAVROS
- Sent on every behaviour change and resent every 30 s for late-connecting ground stations

Result:

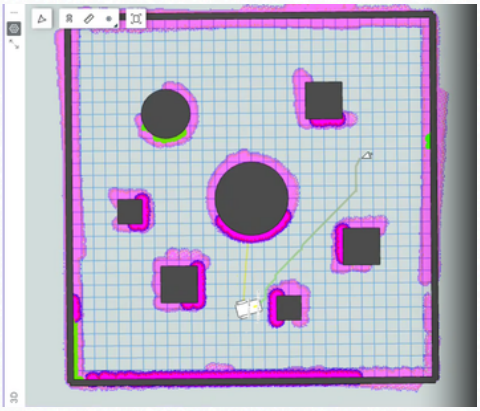
- Verified end to end in ArduPilot SITL and Mission Planner
- Received Takeoff (S,2), FlyToWaypoint (S,8), RepeatLap (S,15) and WaitForReconComplete (S,18) during a simulated mission

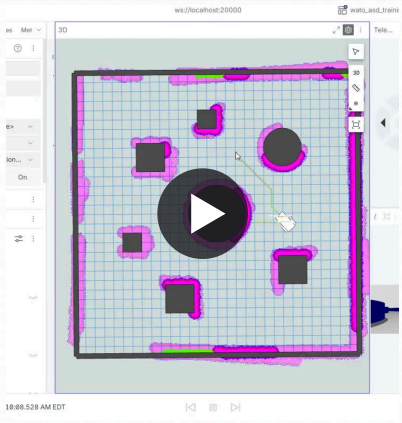
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AUTONOMOUS ROBOT NAVIGATION AND LIDAR MAPPING - C++, ROS 2, FOXGLOVE







What:

- A four-node navigation stack that drives a simulated robot to a chosen goal around static obstacles
- Built in a group of three for the WATonomous admission assignment & seperately built a visual demo

How:

- Turned 256-beam LiDAR scans into a 0.1 m costmap with 0.65 m obstacle inflation
- Merged local costmaps into a 100 × 100 m global map as the robot moved
- Planned with 8-direction A* and followed the path with Pure Pursuit at 10 Hz (1.1 m lookahead)

Result:

- Robot reached goals on both sides of the arena's large central obstacle in Gazebo, verified in Foxglove
- Path replanned as new obstacles were mapped

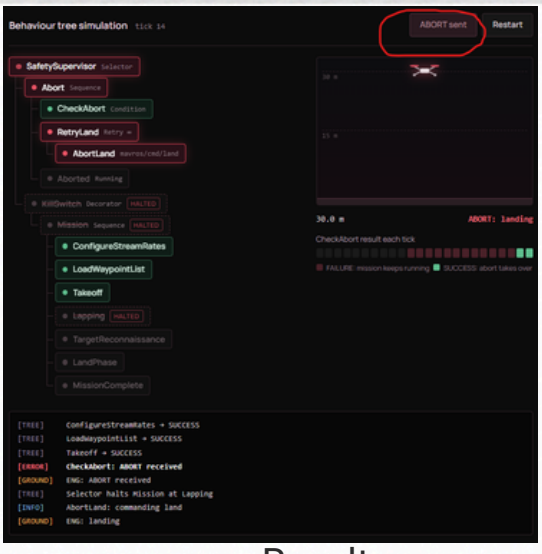
EMERGENCY ABORT SYSTEM - PYTHON, ROS 2, MAVROS

```
SafetySupervisor          Selector: re-checks from the top every tick
├─ Abort                  Sequence
│  ├─ CheckAbort          FAILURE until ABORT arrives, then SUCCESS
│  ├─ RetryLand           resends the land call if it is rejected
│  │  └─ AbortLand        calls mavros/cmd/land, waits for disarm
│  └─ Aborted             RUNNING forever, so the mission never resumes
└─ KillSwitch
   └─ Mission             runs only while Abort fails
      ├─ ConfigureStreamRates
      ├─ LoadWaypointList
      ├─ Takeoff
      ├─ Lapping
      ├─ TargetReconnaissance
      ├─ LandPhase
      └─ MissionComplete
```

```
pside-1  | [manager-9] [WARN] [1792650785.221501840] [engine_manager]: ConfigureStreamRates: waiting
r MAVROS to connect to the FCU
pside-1  | [manager-9] [WARN] [1792650790.927086939] [engine_manager]: ConfigureStreamRates: waiting
r MAVROS to connect to the FCU
pside-1  | [manager-9] [ERROR] [1792650794.726913702] [engine_manager]: CheckAbort: ABORT received
pside-1  | [manager-9] [INFO] [1792650795.325618592] [engine_manager]: to ground: ENG: ABORT received
pside-1  | [manager-9] [WARN] [1792650796.527381194] [engine_manager]: AbortLand: waiting for 'mavros
tate'
pside-1  | [manager-9] [WARN] [1792650801.958886014] [engine_manager]: AbortLand: waiting for 'mavros
tate'
pside-1  | [manager-9] [WARN] [1792650807.402322226] [engine_manager]: AbortLand: waiting for 'mavros
tate'
pside-1  | [mavros_node-1] [INFO] [1792650812.369052262] [mavros.mavros_router]: link[1000] trying to
econnect...
pside-1  | [mavros_node-1] [ERROR] [1792650812.370137883] [mavros.mavros_router]: link[1000] reconnec
ailed: DeviceError:tcp:connect: Connection refused
```

Abort not sent





What:

- An emergency abort for WARG's multi-drone system, where several drones fly in different directions at once, so an operator can stop any drone that isn't moving safely
- Checks the ground station stream, the same one that sends coordinate commands, at the start of every behaviour tree cycle. On ABORT, it switches the drone to LAND mode and stops the behaviour tree from continuing the mission

How:

- Added a Selector above the mission so the abort check runs first on every behaviour tree tick
- CheckAbort latches on the first message to /commander/abort, then the Selector halts the mission branch
- AbortLand calls MAVROS LAND, with automatic retry if the flight controller rejects it

Result:

- Bench test: three ABORT messages triggered CheckAbort: ABORT received and ENG: ABORT received, and the tree switched to AbortLand